

# Markov Chains — Memoryless Step-by-Step Worlds

In this model, chance of each event depends only on previous event. This key feature, called Markov property or memorylessness, means future outcomes depend only on current situation, not on what happened before — history irrelevant given present. This unique memoryless quality allows movement between states based on set probabilities. Markov chains are useful for understanding systems that change step by step, with each step based on current situation. They provide basis for many predictions, simulations, and algorithms used in various fields like science, engineering, and daily life. By using Markov chains, people can spark innovation and gain deeper understanding in many areas.

Formally, stochastic process  $\{X_n\}_{n \geq 0}$  satisfies Markov property:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i) = P_{ij}$$

No need to remember path —  $P_{ij}$  suffices.

## Core Concepts:

- **State Space ( $\Omega$ ):** Set of all possible "states" or conditions system can be in. Discrete — like "sunny" vs "rainy", or letters A-Z — or continuous — e.g., temperature. For Markov chain we usually assume finite or countable  $\Omega = \{1, \dots, N\}$ .
- **Transitions:** Movement from one state to another at each step — often representing unit of time. Sequence  $X_0 \rightarrow X_1 \rightarrow X_2 \dots$  random walk on state graph.
- **Transition Matrix ( $P$ ):** Square matrix where entry  $P_{ij}$  represents probability of moving from state  $i$  to  $j$  in one step. Each row sums to 1 — stochastic matrix —  $P_{ij} \geq 0, \sum_j P_{ij} = 1$ .  $n$ -step transitions  $P^n$  — Chapman-Kolmogorov:  $P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}$  — matrix multiplication.
- **Stationary Distribution ( $\pi$ ):** Long-term "steady state" where probability of being in any given state remains constant even as transitions continue. Mathematically,  $\pi P = \pi, \sum_i \pi_i = 1, \pi_i \geq 0$  — left eigenvector with eigenvalue 1. If chain irreducible — can get from any state to any other — and aperiodic — no cyclic parity — then  $\pi$  unique and  $P^n \rightarrow \mathbf{1}\pi$  — distribution after many steps converges to  $\pi$  regardless of start — ergodic theorem.  $\pi_j = 1/\mathbb{E}_j[T_j]$  — inverse expected return time.
- **Classification:** States recurrent if  $P(\text{return infinitely often}) = 1$ , transient otherwise. Recurrent positive if expected return finite — yields stationary mass. Absorbing if  $P_{ii} = 1$  — chain stops.

## Smartphone Text Prediction

The keyboard that suggests next word — "I love you \_\_\_\_"  $\rightarrow$  "so much / you too" — is direct application of Markov chains, evolved into n-grams and neural language models.

### 1. The problem as Markov chain:

Text is stochastic process over vocabulary  $V$  — state space  $\Omega = V$  size ~50k words plus punctuation. We want  $P(X_{n+1} \mid X_n, \dots, X_0)$ . Exact history huge —  $V^n$  possibilities — impossible. Markov assumption approximates by limited memory.

- **Unigram model (0th-order Markov):**  $P(w_{n+1})$  independent of history — just word frequency. Predicts "the" always most likely — poor.
- **Bigram model (1st-order Markov):**  $P(w_{n+1} \mid w_n)$  — chance of each word depends only on previous word. Transition matrix  $P$  size  $V \times V$  —  $P_{ij} = P(w_j \mid w_i) = \text{Count}(w_i w_j) / \text{Count}(w_i)$ . This is exactly Markov chain where state = previous word.

- **Trigram model (2nd-order Markov):**  $P(w_{n+1} \mid w_n, w_{n-1})$  — depends on two previous words. Can be seen as Markov chain on expanded state space  $\Omega = V^2$  — state = pair  $(w_{n-1}, w_n)$  — transitions to  $(w_n, w_{n+1})$  with prob  $P(w_{n+1} \mid w_{n-1}, w_n)$ . General  $n$ -gram =  $(n - 1)$ -order Markov chain.

Smartphone keyboards classically used up to trigram or 4-gram.

## 2. Training — Counting on your phone:

Your phone builds personalized transition matrix from:

- Large background corpus — e.g., web, Wikipedia — to estimate general English  $P(w_j \mid w_i)$ .
- Your typing history — on-device learning — to personalize.

Estimation simple maximum likelihood:

$$P_{\text{MLE}}(w_n \mid w_{n-1}) = \frac{C(w_{n-1}, w_n)}{C(w_{n-1})}$$

where  $C$  counts in corpus. Example: if you typed "good morning" 20 times and "good night" 5 times,  $C(\text{good}) = 25$ , so  $P(\text{morning} \mid \text{good}) = 0.8$ ,  $P(\text{night} \mid \text{good}) = 0.2$ .

Because many bigrams never seen —  $V^2 = 2.5B$  entries sparse — smoothing needed: add- $\alpha$ , backoff to unigram, Kneser-Ney smoothing — if "San Francisco" never seen, fall back to  $P(\text{Francisco})$  small but non-zero, avoid zero probabilities.

On-device personalization uses same counts but weighted: your "prof\_tmch" vocabulary — e.g., math terms — gets higher probability.

## 3. Prediction — Inference:

When you type "I am going to", keyboard looks up state = "to" for bigram, or ("going","to") for trigram, reads row  $P_{\text{state}, \cdot}$  — distribution over next word — sorts top 3 with highest probability and displays.

Formally, prediction =  $\arg \max_w P(w \mid \text{context})$ . With stationary distribution, long-run text generated by sampling repeatedly from  $P$  converges to style of corpus.

Example walk: State "happy" → transition row: birthday 0.3, to 0.2, hour 0.1... Suggest "birthday". If you select, new state "birthday", next row suggests "party", etc.

This is why Markov text sometimes produces locally plausible but globally incoherent sentences — because memory only 1-2 words, it forgets earlier subject — e.g., "The cat sat on the mat and the mat was very happy birthday" — plausible bigrams but no long memory.

## 4. Beyond Markov — From n-grams to Transformers:

Modern keyboards — Gboard, iOS — now use neural extensions but still Markov-inspired:

- **Hidden Markov Model (HMM):** Used for typo correction — you observe noisy keystroke  $Y_n$  — "teh" — hidden true state  $X_n$  — "the" — model  $P(Y \mid X)$  emission +  $P(X_n \mid X_{n-1})$  transition, decode most likely hidden sequence via Viterbi algorithm.

- **LSTM / Transformer:** Generalization where state is not just last word but learned continuous vector  $h_n$  summarizing history —  $P(w_{n+1} | h_n)$  — still Markov property in  $h$ -space:  $h_{n+1} = f(h_n, w_{n+1})$ . So deep learning = Markov chain with huge learned state space — 4096-dim vector — vs discrete  $V^2$ .

But core idea unchanged from Markov 1906 analyzing Pushkin: future word depends only on finite recent context summarized in current state, transition matrix learned from counts, stationary distribution = language's long-run word frequencies.

Thus every time your phone suggests "you" after "love", it's performing one step of a Markov chain trained on billions of sentences —  $P_{ij}$  lookup — direct descendant of Andrey Markov counting vowels in Eugene Onegin.

## Diffeomorphism — Smooth Perfect Pairing

In differential geometry, diffeomorphism is the notion of "same" — two manifolds are same smooth shape if there is smooth invertible map with smooth inverse between them — isomorphism in category of smooth manifolds — analogous to isomorphism of groups, homeomorphism of topological spaces, but stronger than homeomorphism, requiring calculus not just continuity.

Intuition: If you have two shapes made of infinitely stretchable but non-tearable rubber that also preserves smoothness — no creasing — diffeomorphism is reparameterization that shows they are identical as smooth objects.

### The Three Requirements

For function  $f : M \rightarrow N$  between two smooth manifolds — e.g.,  $M, N$  are surfaces, curves,  $\mathbb{R}^n$ , Lie groups — to be diffeomorphism, must satisfy three conditions:

1. **Bijective:** It is perfect 1-to-1 pairing; every point on first shape maps to exactly one point on second, and vice versa —  $f$  invertible as set map — there exists inverse map  $f^{-1} : N \rightarrow M$  as function — no collapsing, no missing points.
2. **Differentiable ( $C^k$  or  $C^\infty$ ):** Function is smooth. If you move along first shape, corresponding movement on second shape changes smoothly, with no sudden jumps or sharp turns. Formally, in local charts  $\phi : U \subset M \rightarrow \mathbb{R}^m$ ,  $\psi : V \subset N \rightarrow \mathbb{R}^n$ , composition  $\psi \circ f \circ \phi^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$  is  $k$ -times continuously differentiable — usually  $C^\infty$  — infinitely differentiable, or  $C^r$  —  $r$  derivatives. All partial derivatives exist and continuous. This ensures tangent vectors push forward — differential  $df_p : T_p M \rightarrow T_{f(p)} N$  linear map exists at each point.
3. **Inverse is Differentiable:** The "return trip"  $f^{-1} : N \rightarrow M$  must also be smooth —  $C^\infty$  in charts. This is crucial part that distinguishes it from standard smooth bijection — smooth bijection can have non-smooth inverse — e.g.,  $f(x) = x^3 : \mathbb{R} \rightarrow \mathbb{R}$  smooth, bijective, but inverse  $f^{-1}(y) = y^{1/3}$  not differentiable at 0 — derivative infinite — so not diffeomorphism. Requires  $df_p$  invertible at every  $p$  — by Inverse Function Theorem, if  $df_p$  isomorphism of tangent spaces and  $f$  bijective, then locally smooth inverse exists; global bijectivity plus  $df_p$  invertible everywhere  $\implies$  diffeomorphism — if  $M, N$  same dimension — condition  $\det(Df) \neq 0$  everywhere.

Together: diffeomorphism = smooth isomorphism, smooth structure preserving.

If  $M = N$ , diffeomorphism  $f : M \rightarrow M$  called self-diffeomorphism — group  $\text{Diff}(M)$  — infinite-dimensional Lie group — Lie algebra = vector fields  $\mathfrak{X}(M)$  — flows.

### Diffeomorphism vs. Homeomorphism — Smooth vs Topological Same

While they sound similar, difference is about "tools" allowed — regularity:

- **Homeomorphism (Topology):** Cares about connectivity, continuity. Map  $f$  bijective, continuous, inverse continuous —  $C^0$  isomorphism. As long as you don't tear object — cut — or glue distinct points, it's same. Stretching, bending allowed arbitrarily, even non-differentiable kinks allowed as long as continuous. Topological invariants preserved: number of components, holes, compactness, fundamental group. Example: square is homeomorphic to circle — map radial projection:  $(x, y) \mapsto (x, y) / \max(|x|, |y|)$  times radius? Actually circle to square via  $L^\infty$  norm — continuous both ways, corners smoothed topologically but not smoothly. Also coffee mug homeomorphic to donut — genus 1.
  - **Diffeomorphism (Differential Geometry):** Cares about calculus — smooth structure. Need transition to be smooth — derivatives preserved. Preserves not just topology but differentiable structure: tangent spaces, ability to do calculus, curvature, differential forms. Invariants: dimension, orientability as smooth, de Rham cohomology, smooth characteristic classes. Example: square is *not* diffeomorphic to circle as submanifolds of  $\mathbb{R}^2$  with induced smooth structure? Actually as abstract manifolds with boundary vs without boundary? Let's clarify: boundary square — manifold with corners — interior? Usually circle  $S^1$  smooth 1-manifold, boundary of square — topological 1-manifold but not smooth submanifold at corners — its inclusion map not smooth immersion at corners — derivative jumps. There exists abstract diffeomorphism between square boundary as topological circle with smoothed corners? As abstract smooth manifolds, any topological circle can be smoothed to be diffeomorphic to standard  $S^1$  — can round corners via smooth reparameterization. So square *loop* with corners considered as subset of  $\mathbb{R}^2$  is not *smooth submanifold* — embedding not smooth — but as abstract 1-manifold it's diffeomorphic to  $S^1$  after reparameterization that slows down near corners. Better example: cube vs sphere in  $\mathbb{R}^3$ : cube surface not diffeomorphic as submanifold to sphere because not smooth at edges, though homeomorphic; but as abstract manifold, after smoothing corners via radial map  $x \mapsto x/|x|$ , cube surface is diffeomorphic to  $S^2$  — map is smooth? Radial projection from cube to sphere is diffeomorphism except at edges? Actually radial projection cube  $\rightarrow$  sphere is homeomorphism,  $C^0$  but not  $C^1$  at edges — derivative discontinuous — but there exists *different* diffeomorphism that smooths edges — can round cube.
- Sharp distinction: In dimensions  $\leq 3$ , homeomorphic smooth manifolds are diffeomorphic — topological classification = smooth classification — but in higher dimensions exotic phenomena: Milnor 1956 discovered exotic 7-spheres — topological  $S^7$  but not diffeomorphic to standard  $S^7$  — 28 distinct smooth structures on  $S^7$ . So homeomorphic does NOT imply diffeomorphic in general — smooth structure extra data.  $\mathbb{R}^4$  has uncountably many exotic smooth structures — exotic  $\mathbb{R}^4$ s homeomorphic but not diffeomorphic to standard  $\mathbb{R}^4$  — only dimension 4 has this.

## Why diffeomorphism matters:

- General relativity: spacetime manifolds considered up to diffeomorphism — diffeomorphism invariance — Einstein equations covariant under  $\text{Diff}(M)$  — physics same after smooth coordinate change — active diffeomorphism moves points.
- Lie theory: Lie group  $G$  acts on itself by diffeomorphisms — left multiplication  $L_g : h \mapsto gh$  diffeomorphism — Lie algebra = left-invariant vector fields.
- Dynamical systems: Two systems topologically conjugate vs smoothly conjugate — smooth conjugacy preserves eigenvalues of fixed points — Hartman-Grobman —  $df$  matters.
- Hamiltonian mechanics: Symplectomorphism — diffeomorphism preserving symplectic form  $\omega$  —  $f^*\omega = \omega$  — preserves Hamiltonian structure — stronger than diffeomorphism.

In short: Homeomorphism says you can deform without tearing — topology. Diffeomorphism says you can deform without tearing *and* without creasing — smooth geometry. Square vs circle: topologically same, but as embedded shapes with sharp corners, not smoothly same via inclusion — need smoothing to become diffeomorphic — captures intuition that calculus cares about corners while topology does not.

# Ergodicity — When Time Average Equals Space Average

Ergodicity is the formal bridge between what we *observe* over long time for a single trajectory and what we *predict* on average across many possible states. It answers: does a single particle wandering forever see the whole space in the correct proportions?

Intuitively, a system is ergodic if it cannot be split into two invariant pieces of positive size — any trajectory, except for a negligible set, explores the entire phase space — it forgets its initial condition, not by converging to a point, but by equidistributing.

In short: Ergodicity is the hypothesis that makes statistical mechanics, MCMC, and steady-state simulation possible — one long run suffices to learn global statistics — when it fails, we need many runs or longer timescales, revealing hidden structure and phase transitions.

## Historical Origin

Term coined by Boltzmann in 1884 from Greek *ergon* — work + *odos* — path — in kinetic theory of gases. Boltzmann's ergodic hypothesis: a single gas molecule over infinite time visits every point on constant-energy surface — phase space — with frequency proportional to volume — justifying replacing time averages in lab with ensemble averages in statistical mechanics — Maxwell-Boltzmann distribution.

Original hypothesis false in literal sense — continuous curve cannot fill  $(6N - 1)$ -dimensional surface — measure zero — but von Neumann 1932 and Birkhoff 1931 proved rigorous statistical versions that saved the idea — mean and pointwise ergodic theorems — foundation of ergodic theory.

## Formal Definition — Three Equivalent Views

Let  $(X, \mathcal{B}, \mu)$  be probability space —  $X$  phase space,  $\mu(X) = 1$  — and  $T : X \rightarrow X$  measure-preserving transformation:  $\mu(T^{-1}A) = \mu(A)$  for all measurable  $A$  — dynamics conserves volume — e.g., Hamiltonian flow preserves Liouville measure, or Markov chain preserves stationary distribution  $\pi$  where  $\pi P = \pi$ .

$T$  is **ergodic** if any of following equivalent holds:

**1. Indecomposability:** Every invariant set is trivial. If  $T^{-1}A = A \bmod \text{measure}$  — i.e.,  $\mu(A \Delta T^{-1}A) = 0$  — then  $\mu(A) = 0$  or 1. No non-trivial subset trapped by dynamics — cannot split  $X = A \sqcup A^c$  into two invariant pieces of positive measure.

**2. Pointwise Ergodic Theorem — Birkhoff 1931:** For any integrable observable  $f \in L^1(\mu)$ ,

$$\bar{f}(x) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \int_X f d\mu$$

for  $\mu$ -almost every  $x$ . Time average along trajectory  $T^n x$  equals space average over  $\mu$ . Convergence almost everywhere and in  $L^1$ .

**3. Mean Ergodic Theorem — von Neumann 1932:** Same limit holds in  $L^2$  sense:  $\|\frac{1}{N} \sum_{n=0}^{N-1} U^n f - \int f d\mu\|_{L^2} \rightarrow 0$ , where  $Uf = f \circ T$  Koopman operator — unitary on  $L^2$ .

If system not ergodic, limit exists but equals conditional expectation onto invariant sigma-algebra — average over ergodic components — not global average.

## Intuition and Non-Examples

- **Non-ergodic:** Identity map  $T(x) = x$  — every set invariant — time average =  $f(x)$  — initial value — never forgets — space average =  $\int f$ . Fails unless space trivial. Rotation of circle by rational angle  $T_\alpha(x) = x + p/q \bmod 1$  — orbit finite, visits  $q$  points only — not dense — set  $A = q$  equally spaced arcs invariant? Actually  $T^q = id$ , invariant sets many.
- **Ergodic:** Irrational rotation  $T_\alpha(x) = x + \alpha \bmod 1$ ,  $\alpha \notin \mathbb{Q}$  — every orbit dense, equidistributed — Weyl's criterion — time average of interval = its length. But not mixing — returns regularly.
- **Ergodic and mixing:** Doubling map  $T(x) = 2x \bmod 1$ , Bernoulli shift, Arnold cat map  $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  on torus  $\mathbb{T}^2$  — stretches and folds — preserves Lebesgue. Stronger chaos.

Hierarchy of randomness — measure-preserving systems classified:

$$\text{Bernoulli} \implies \text{mixing} \implies \text{weak-mixing} \implies \text{ergodic}$$

**Mixing** is stronger:  $\mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$  — asymptotically independent — any set  $A$  smeared uniformly over space after long time — like drop of ink in water — while ergodicity only requires Cesàro average of that to converge. Irrational rotation ergodic but not mixing — intervals return.

**Weak-mixing:**  $\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| \rightarrow 0$  — no non-trivial eigenfunctions.

## Markov Chain Ergodicity

For Markov chain with transition matrix  $P$  and stationary distribution  $\pi$ , ergodicity corresponds to **irreducibility + aperiodicity**:

- **Irreducible:** Can get from any state  $i$  to any  $j$  in some steps — single communicating class — equivalent to indecomposability — no invariant subset  $A$  with  $P_{ij} = 0$  for  $i \in A, j \notin A$ .
- **Aperiodic:** Greatest common divisor of return times to state  $i$  is 1 — no bipartite cycling.

If both hold — chain ergodic — then:

$$\lim_{n \rightarrow \infty} P_{ij}^n = \pi_j$$

independent of  $i$ , and

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{\{X_n=j\}} = \pi_j \quad \text{a.s.}$$

— fraction of time spent in  $j$  converges to stationary probability.

This is why PageRank works: web graph made irreducible and aperiodic by teleportation —  $P' = \alpha P + (1 - \alpha)\mathbf{1}v^T$  — ensures unique stationary  $\pi$  — PageRank vector — and random surfer's time average = PageRank.

If periodic — e.g., simple random walk on bipartite graph —  $P^n$  does not converge — oscillates even-odd — but Cesàro average still converges to  $\pi$  — ergodic but not mixing in Markov sense. Aperiodic needed for mixing.

## Why Physics Cares — Foundation of Statistical Mechanics

Statistical mechanics assumes we can compute macroscopic observables — pressure, temperature — as ensemble averages  $\int f d\mu_{\text{microcanonical}}$  over energy surface, but experiment measures time average over single trajectory of  $10^{23}$  particles. Ergodicity justifies equality — if Hamiltonian flow ergodic on energy surface, time average = space average for almost every initial condition — measurement of one system over long time equals average over many copies at one time.

Boltzmann's program: Show gas dynamics — hard spheres in box — ergodic. Mathematically extremely hard — KAM theory shows many Hamiltonian systems *not* ergodic — integrable systems have invariant tori that block ergodicity — e.g., solar system. Sinai billiard — particle bouncing in square with convex obstacle — proved ergodic and mixing 1970 — first rigorous gas-like system.

Even when not fully ergodic, system may be **ergodic on components** — phase space splits into ergodic components — each with own microcanonical measure — metastable states — glasses, folded proteins.

## Examples of Ergodic vs Non-Ergodic Systems

This section makes the definition concrete — when does time average = space average, and when does it fail catastrophically?

### *Ergodicity in Stationary Stochastic Processes — When One Trajectory Is Enough*

For stationary stochastic process  $\{X(t)\}$  with invariant measure  $\mu$ , ergodicity means we can estimate statistics from single sample path — e.g., mean  $\mu = \mathbb{E}[X(t)]$  estimated by time average

$$\hat{\mu}_T = \frac{1}{T} \int_0^T X(t) dt \rightarrow \mu \quad \text{a.s. as } T \rightarrow \infty$$

— Birkhoff.

When does this hold? Requires autocorrelation  $R(\tau) = \text{Cov}(X(t), X(t + \tau))$  decays fast enough —  $\frac{1}{T} \int_0^T R(\tau) d\tau \rightarrow 0$  as  $T \rightarrow \infty$ . If  $R(\tau)$  does not decay, process remembers forever, time average fluctuates.

### **Example: Ornstein-Uhlenbeck process — ergodic archetype:**

$$dX_t = -\theta X_t dt + \sigma dB_t$$

— mean-reverting — spring pulling to 0 with strength  $\theta$ , plus Brownian kicks. This is velocity of particle with friction — Langevin equation. Solution:

$$X_t = X_0 e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dB_s$$

Stationary distribution  $N(0, \sigma^2/2\theta)$  — Gaussian with variance  $\sigma^2/(2\theta)$  — exists, unique, invariant. Autocorrelation exponential  $R(\tau) = \frac{\sigma^2}{2\theta} e^{-\theta|\tau|}$  — decays, integrable. So ergodic and mixing — even  $L^2$  convergence with rate  $\theta$ . If you watch one particle long enough, histogram of positions converges to Gaussian  $N(0, \sigma^2/2\theta)$  — time average = ensemble.

**Non-example: Brownian motion itself not ergodic.** Why? Not stationary — variance  $\text{Var}(B_t) = t$  grows without bound — no invariant probability measure —  $\mu$  would be Lebesgue infinite measure, not normalizable. Time average  $\frac{1}{T} \int_0^T B_t dt$  does not converge to constant — scales like  $\sqrt{T}$  random. However, *increments*  $X_t = B_{t+1} - B_t$  are stationary and ergodic — i.i.d.  $N(0, 1)$ . Similarly,  $Y_t = B_{t+1} - B_t$ ? Ergodic. So Brownian motion is null-recurrent — returns infinitely often but expected return time infinite — no stationary probability.

Other ergodic examples: finite irreducible aperiodic Markov chain — random walk on connected non-bipartite graph — positive recurrent; irrational rotation  $x \mapsto x + \alpha \bmod 1$  — Lebesgue measure ergodic; doubling map  $x \mapsto 2x \bmod 1$  — mixing.

### **Breaking Ergodicity — When System Gets Trapped — Glasses, Learning, Finance**

Many interesting real systems **break ergodicity**: time average  $\neq$  ensemble average — system trapped in component, phase space fractures. Formal definition: existence of invariant sets  $0 < \mu(A) < 1$  — decomposable.

- **Spin glasses — physics canonical example:** Below critical temperature  $T_c$ , free energy landscape fractures into exponentially many valleys — pure states — separated by barriers  $\Delta E \sim N$  — system size. Dynamics — Metropolis — stuck in one valley for time  $\tau \sim \exp(\Delta E/T)$  astronomical for  $N = 10^23$ . Time average over simulation  $10^9$  steps = average within valley — not Boltzmann average over all valleys. Ensemble average over many replicas at  $t = 0$  samples all valleys. Ergodicity restored only as  $T \rightarrow \infty$  or time  $\rightarrow \exp(N)$ . Parisi's replica symmetry breaking describes decomposition.
- **Machine learning — SGD as random dynamical system:** Loss landscape  $L(\theta)$  highly non-convex. SGD  $d\theta = -\nabla L(\theta)dt + \sqrt{2\eta}\Sigma^{1/2}dB_t$  — small learning rate  $\eta$  corresponds to low temperature — chain not ergodic over landscape — gets stuck in local minima, basin of attraction. Time average of weights along training = point estimate, not posterior average. Need annealing — increase  $\eta$ , Langevin dynamics — or large-batch noise to restore ergodicity and explore. Modern deep learning intentionally operates in non-ergodic regime — early stopping picks one basin.
- **Finance — geometric Brownian motion and ergodicity economics:**  $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma B_t)$  — price of stock with drift  $\mu$ , vol  $\sigma$ . Ensemble average  $\mathbb{E}[S_t] = S_0 e^{\mu t}$  grows at rate  $\mu$  — average over many parallel universes. But median trajectory  $S_0 e^{(\mu - \sigma^2/2)t}$  — typical single trajectory — growth rate  $\mu - \sigma^2/2$ . If  $\sigma^2 > 2\mu$ , ensemble grows to infinity while almost every individual goes to zero! Time average growth rate

$$g = \lim_{T \rightarrow \infty} \frac{1}{T} \log \frac{S_T}{S_0} = \mu - \frac{\sigma^2}{2} \quad \text{a.s.}$$

— not  $\mu$ . This gap drives "ergodicity economics" — Peters 2019 — argues expected utility theory uses ensemble average — wrong for individual who lives one trajectory — should use time-average growth — Kelly criterion  $f^* = \mu/\sigma^2$  maximizes time-average, not expected wealth. Lottery, insurance, leverage decisions flip when using time average.

**Testing ergodicity in data:** Given time series, split into  $K$  windows, compute time averages  $\hat{\mu}_T^{(k)}$ . If ergodic,  $\hat{\mu}_T^{(k)} \rightarrow \mu$  same for all  $k$  as  $T$  large — variance across windows  $\rightarrow 0$ . If non-ergodic, variance persists — different initial conditions lead to different long-run averages — e.g., cell differentiation: genetically identical cells have different steady-state protein levels — lineage memory. Many biological systems show ergodicity breaking — single-cell gene expression bimodal.

### ***Russian Roulette — The Classic Non-Ergodic Gambler's Ruin***

In ergodic system, average of group at one point in time same as average of one person over long period. Russian Roulette is **non-ergodic** because "death" is absorbing state that stops process — once entered, cannot leave — invariant set.

Model: state space  $\Omega = \{\text{Alive with \$0, Alive with \$1M, Dead}\}$ .  $T$ : with prob 5/6 go to Alive with \$1M, with prob 1/6 go to Dead — absorbing:  $P(\text{Dead} \rightarrow \text{Dead}) = 1$ .

1. **Ensemble Average — The Group View — What happens to many people at once:** If 6 people play single round simultaneously, expected value for group:

$$P(\text{Survival}) = 5/6, \quad P(\text{Death}) = 1/6$$

Prize = \$1,000,000 if survive, 0 if dead — assume wealth zero after death.

Average wealth of group:

$$E = (5/6 \times \$1,000,000) + (1/6 \times \$0) = \sim \$833,333$$

Result: "Average" person in this group is millionaire — ensemble suggests favorable game — expected value positive.



2. **Time Average — The Individual View — What happens to one person over time:** If one person plays 6 rounds in a row, probability surviving  $n$  rounds:

$$P_s(n) = (5/6)^n$$

For 6 rounds:  $P_s(6) = (5/6)^6 \approx 0.334$  — only 33% survive 6 games. Probability of death  $P_d = 1 - 0.334 = 0.666$  — 66.6%.

As  $n \rightarrow \infty$ ,  $P_s(n) \rightarrow 0$  exponentially fast. Expected number of rounds until death geometric mean  $1/(1/6) = 6$ . Time-averaged wealth along infinite trajectory: eventually hits Dead, stays 0 forever — average  $\rightarrow 0$  — not \$833k.

3. **The Ergodicity Gap:** System non-ergodic because Ensemble Average  $\neq$  Time Average. While group looks successful — 5/6 alive millionaires — individual eventually hits absorbing state where wealth irrelevant. Mathematically, invariant measure concentrated on Dead —  $\mu(\{\text{Dead}\}) = 1$  — only stationary distribution, but transient phase shows high average — metastable.

This is why insurance, risk management focus on time average — you cannot average across parallel universes where you died — you live one trajectory, absorbing states dominate long-run. Same logic: driving without seatbelt, Russian Roulette finance — leveraged bets with risk of ruin.

### ***A Deck of Cards — Huge State Space, But Ergodic Shuffling***

Number of ways to arrange 52-card deck is  $8.06 \times 10^{67} = 52!$  — classic illustration of combinatorial explosion and why random shuffling is effectively unique, and also example of ergodic Markov chain on symmetric group.

**To understand why number so large and repeat virtually impossible, we look at Fundamental Counting Principle — rule of product:**

If there are  $n$  ways to do one thing and  $m$  ways to do another, there are  $n \times m$  ways to do both — when choices independent.

- If task broken into stages — event 1, event 2... — total ways = product of choices at each stage.
- Independent Events: formula works when selection in one step does not affect number of options in another? Actually for permutations, dependence: remaining cards decrease — still product.
- Formula: Total Outcomes =  $M_1 \times M_2 \times \dots \times M_n$ .
- Application: Used extensively in probability and combinatorics — outfits, passwords.

**Calculate total permutations:** When you build deck card by card, choices decrease by one:

For first card, 52 choices, second 51 remaining, third 50, etc.

Total unique arrangements:

$$52 \times 51 \times 50 \times \dots \times 3 \times 2 \times 1 = 52!$$

Exactly:

80, 658, 175, 170, 943, 878, 571, 660, 636, 856, 403, 766, 975, 289, 505, 440, 883, 277, 824, 000, 000, 000, 000

**Compare to human history — Ergodicity timescale:** To see if humans could have repeated shuffle by chance, estimate total shuffles ever performed. Even using extremely generous assumptions:



Corollary: Almost all reals are uncomputable, transcendental, undefinable — countable sets: computable numbers, algebraic numbers, definable numbers — each countable, so their complement in  $\mathbb{R}$  has cardinality  $\mathfrak{c}$  — majority of reals never describable.

### How Big is $\mathfrak{c}$ ?

Mathematically, cardinality of continuum equal to  $2^{\aleph_0}$  — 2 raised to power "aleph-null".

- $\aleph_0$  — Aleph-null: size of natural numbers — countable infinity —  $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$  — integers, fractions — all countable — can enumerate —  $|\mathbb{Q}| = \aleph_0$  because  $p/q$  map to  $\mathbb{N}^2$  countable.
- $2^{\aleph_0}$ : size of power set of natural numbers —  $\mathcal{P}(\mathbb{N})$  — set of all subsets of  $\mathbb{N}$  — each subset corresponds to infinite binary sequence — characteristic function — which corresponds to real in binary expansion. So  $|\mathcal{P}(\mathbb{N})| = |\{0, 1\}^{\mathbb{N}}| = 2^{\aleph_0} = \mathfrak{c}$ . Also  $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$ .
- Interestingly, number of points on 1-inch line segment exact same as number of points in entire universe or 3D cube —  $|[0, 1]| = |\mathbb{R}^3| = \mathfrak{c}$ . Cantor proved  $|\mathbb{R}^n| = \mathfrak{c}$  for any finite  $n$  — bijection between line and plane —  $|\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$  — e.g., interleave decimal digits:  $0.a_1a_2\dots, 0.b_1b_2\dots \mapsto 0.a_1b_1a_2b_2\dots$  — after handling 0.999... issues. So dimension does not increase cardinality — topological dimension  $\neq$  cardinal size. Even  $\mathbb{R}^{\mathbb{N}}$  — countable product —  $|\mathbb{R}^{\mathbb{N}}| = \mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0} = \mathfrak{c}$  — still continuum. Only power set  $\mathcal{P}(\mathbb{R})$  larger:  $|\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}} > \mathfrak{c}$ .

Cardinal arithmetic:  $\mathfrak{c} + \aleph_0 = \mathfrak{c}$ ,  $\mathfrak{c} \cdot \aleph_0 = \mathfrak{c}$ ,  $\mathfrak{c} + \mathfrak{c} = \mathfrak{c}$ ,  $\mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$  — continuum absorbs countable additions/multiplications.

### The Continuum Hypothesis (CH) — Is There Intermediate Infinity?

Most famous problem in mathematical history, Hilbert's first problem 1900. Asks: Is there any infinity between size of integers  $\aleph_0$  and size of real numbers  $\mathfrak{c}$ ? Formally,  $\mathfrak{c} = \aleph_1$ ? Where  $\aleph_1$  is next cardinal after  $\aleph_0$ .

Define aleph hierarchy:  $\aleph_0 < \aleph_1 < \aleph_2 < \dots$  —  $\aleph_1$  is smallest uncountable cardinal — size of set of countable ordinals.

Question: where does  $\mathfrak{c} = 2^{\aleph_0}$  sit in this hierarchy? CH asserts  $\mathfrak{c} = \aleph_1$  — no intermediate cardinal — continuum is next after countable.

**The Answer — Independence:** Gödel 1940 proved CH consistent with ZFC — built constructible universe  $L$  where CH holds — cannot be disproved. Cohen 1963 proved negation consistent — using forcing — built model of ZFC where  $\mathfrak{c} = \aleph_2$ , or  $\aleph_{17}$ , etc. — you can make continuum arbitrarily large with cofinality  $> \aleph_0$ . Together: CH undecidable using standard set theory ZFC — you can choose to believe there is intermediate size, or believe there isn't, and math remains consistent either way — like parallel postulate in geometry.

This shocked foundations — CH is natural question about real numbers, yet ZFC insufficient to settle.

Generalized Continuum Hypothesis GCH:  $2^{\aleph_\alpha} = \aleph_{\alpha+1}$  for all  $\alpha$ .

### ZFC — The Rulebook Where Question Lives

Zermelo-Fraenkel set theory with Axiom of Choice — ZFC — standard foundational system for modern mathematics, designed to avoid paradoxes like Russell's — set of all sets not containing themselves — by defining sets through axioms. It defines sets via single membership relation  $\in$ , building structures from empty set to define complex math objects — numbers as von Neumann ordinals:  $0 = \emptyset$ ,  $1 = \{\emptyset\}$ ,  $2 = \{\emptyset, \{\emptyset\}\}$  etc.

### Core Axioms of ZFC — Building Universe $V$ :

ZFC is not just list of rules — it is construction manual for the cumulative hierarchy  $V$  — the universe of all sets. Start with nothing, iterate operations, avoid paradoxes. Each axiom corresponds to one closure operation.

Picture:  $V_0 = \emptyset$ ,  $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ ,  $V_\lambda = \bigcup_{\alpha < \lambda} V_\alpha$  for limit  $\lambda$ . Then  $V = \bigcup_{\alpha \in Ord} V_\alpha$ . Axioms guarantee each step exists and that  $V$  is well-founded.

### Extensionality — Identity Criterion

Two sets equal iff they have same elements:  $\forall x (x \in A \iff x \in B) \implies A = B$  — sets determined by members.

What it does: defines what equality *means* for sets. No hidden intension — set is exactly its extension.  $\{1, 2\} = \{2, 1\} = \{1, 1, 2\}$  — order and multiplicity irrelevant. Without this, membership  $\in$  would not determine identity.

Why needed: Russell's paradox used unrestricted comprehension, but extensionality already in naive set theory — keeps sets as pure collections.

### Empty Set — Starting Point

Exists set  $\emptyset$  containing no elements:  $\exists x \forall y y \notin x$  — starting point.

Formally:  $\exists x \forall y \neg(y \in x)$ . By extensionality, this  $x$  unique — denote  $\emptyset$ .

This is  $V_0$ . Without it, universe could be empty of sets — axioms would be vacuously true. Gives atom from which to build everything:  $0 := \emptyset$  in von Neumann construction.

From empty set alone + other axioms we get all of mathematics — remarkable.

### Pairing — Building Finite Collections

For any sets  $x, y$ , exists set  $\{x, y\}$  — can build unordered pair, then  $\{x\} := \{x, x\}$ .

Axiom:  $\forall x \forall y \exists z \forall w (w \in z \iff w = x \vee w = y)$ .

With Empty Set, we get:  $\{\emptyset\}$ ,  $\{\emptyset, \{\emptyset\}\}$ , etc.

Enables ordered pair — Kuratowski definition:  $(a, b) := \{\{a\}, \{a, b\}\}$  — crucial because order matters for relations, functions.

Pairing + extensionality gives that  $(a, b) = (c, d) \iff a = c \wedge b = d$ .

Without pairing, cannot build set containing two given sets — universe not closed under finite enumeration.

### Union — Flattening

For any set of sets  $F$ , exists set  $\bigcup F$  containing all elements of those sets —  $\bigcup\{\{1, 2\}, \{2, 3\}\} = \{1, 2, 3\}$ .

Formally:  $\forall F \exists U \forall y (y \in U \iff \exists z \in F y \in z)$ .

If  $F = \{A, B\}$ , then  $\bigcup F = A \cup B$  — binary union. So union axiom gives closure under arbitrary — even infinite — unions.

Needed to define  $x \cup \{x\}$  — successor — for Infinity. Also to define  $A \cup B$ , and to build  $V_{\omega+1}$  etc. Without it, sets would be isolated — cannot aggregate members of members.

### Power Set — Exponential Growth

For any set  $x$ , exists set  $\mathcal{P}(x)$  containing all subsets of  $x$  — crucial for generating larger cardinals —  $|\mathcal{P}(x)| = 2^{|x|}$  — from  $\mathbb{N}$  get  $\mathfrak{c}$ .

Formally:  $\forall x \exists y \forall z (z \in y \iff z \subseteq x)$  where  $z \subseteq x := \forall w (w \in z \implies w \in x)$ .

This is the powerhouse — generates hierarchy of infinities. Starting from  $\mathbb{N}$ ,  $\mathcal{P}(\mathbb{N})$  has cardinality  $2^{\aleph_0} = \mathfrak{c} = |\mathbb{R}|$  — continuum. Then  $\mathcal{P}(\mathcal{P}(\mathbb{N}))$  has  $2^{\mathfrak{c}}$ , etc. Cantor's theorem  $|x| < |\mathcal{P}(x)|$  proved using Diagonal — no surjection — so Power Set strictly increases size.

Gives  $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ . Without Power Set, cannot prove existence of  $\mathbb{R}$  — reals as Dedekind cuts — subsets of  $\mathbb{Q}$  — or prove existence of uncountable set. ZF minus Power Set — called  $ZF^-$  — cannot build  $V_{\omega+1}$ .

### ***Infinity — Actual Infinite Exists***

Exists infinite set, used to construct natural numbers —  $\exists I (\emptyset \in I \wedge \forall x \in I \ x \cup \{x\} \in I)$  — gives  $\omega$ .

Formally:  $\exists I (\emptyset \in I \wedge \forall x \in I \ x \cup \{x\} \in I)$  where  $x \cup \{x\}$  uses Pairing + Union.

The set  $I$  is inductive — contains  $0 = \emptyset$ ,  $1 = \{0\} = \{\emptyset\}$ ,  $2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$ , etc. — von Neumann ordinals. Then define  $\omega := \bigcap \{J \subseteq I : J \text{ inductive}\}$  — smallest inductive set — exists by Separation —  $\mathbb{N}$ .

Without Infinity, all sets finite — universe  $V_\omega$  — hereditarily finite sets — model of all other axioms except Infinity — suffices for finitistic math but not for calculus,  $\mathbb{N}$  not a set, only proper class.

Infinity distinguishes potential infinity — arbitrarily large finite — from actual infinity — set containing all naturals as completed totality.

### ***Separation — Subset — Restricted Comprehension — The Paradox Killer***

Subset of existing set can be formed from property  $P(x)$  —  $\{x \in A : P(x)\}$  set — restricted comprehension — avoids Russell's paradox — cannot form  $\{x : x \notin x\}$  without bounding  $A$ .

Axiom Schema — one axiom per formula  $\varphi$ :  $\forall A \exists B \forall x (x \in B \iff x \in A \wedge \varphi(x))$ .

Naive comprehension:  $\exists B \forall x (x \in B \iff \varphi(x))$  for any  $\varphi$  — inconsistent — Russell: take  $\varphi(x) := x \notin x \rightarrow R = \{x : x \notin x\} \rightarrow R \in R \iff R \notin R$  contradiction.

Zermelo's fix 1908: must carve subset out of *already existing* set  $A$ . So  $\{x \in A : x \notin x\}$  exists, but equals  $\{x \in A : x \notin x\}$  — if  $A$  contains  $R$ ,  $R \notin R$ , no contradiction — just  $R \notin B$  — actually  $B$  does not contain itself anyway. Paradox blocked because you cannot form universe of all sets.

Examples: Intersection  $A \cap B = \{x \in A : x \in B\}$ , difference  $A \setminus B$ ,  $\{x \in \mathbb{N} : x \text{ even}\}$ . Defines empty set from any  $A$ :  $\emptyset = \{x \in A : x \neq x\}$ .

Separation alone too weak to build large sets — cannot get  $\bigcup_n \mathcal{P}^n(\mathbb{N})$  — need Replacement.

### ***Replacement — Image of Set Is Set — Transfinite Recursion Engine***

Image of set under definable function also set — if  $F$  definable function and  $A$  set, then  $F[A] = \{F(x) : x \in A\}$  set — allows transfinite recursion, building  $V_{\omega+\omega}$  etc., prevents paradoxes by limiting comprehension.

Schema: If  $\varphi(x, y)$  functional —  $\forall x \in A \exists! y \varphi(x, y)$  — then  $\exists B \forall y (y \in B \iff \exists x \in A \varphi(x, y))$ .

Stronger than Separation — Separation is special case where  $y = x$  and  $\varphi$  is  $x \in A \wedge \psi(x)$ . Replacement says universe closed under definable functions.

Why needed:

- Define sequence  $\omega, \mathcal{P}(\omega), \mathcal{P}(\mathcal{P}(\omega)), \dots$  — map  $n \mapsto \mathcal{P}^n(\omega)$  definable — Replacement collects range  $\{\mathcal{P}^n(\omega) : n \in \omega\}$  into set — then Union gives  $\bigcup_n \mathcal{P}^n(\omega) = V_{\omega+\omega}$ ? Actually need Replacement to get set of iterates.
- Define  $V_\alpha$  for  $\alpha \leq \omega_1$  — transfinite recursion — Replacement guarantees each step stays set.
- Prove existence of  $\aleph_\omega = \bigcup_n \aleph_n$  — need Replacement to collect  $\aleph_n$ .
- Proves Separation not enough — there are models of Z — Replacement + Separation where Replacement fails —  $V_{\omega \cdot 2}$ .

Intuition: If  $A$  small — set — and you map each  $x \in A$  to something — not too many — via definable rule, result not too big — still set, not proper class. Limits size — prevents Burali-Forti paradox of all ordinals.

### Foundation — Regularity — No Infinite Descending Chains

Every non-empty set has  $\in$ -minimal element, prohibiting sets containing themselves —  $x \notin x$  — and ruling out infinite descending membership chains  $x_0 \ni x_1 \ni x_2 \dots$  — ensures universe well-founded,  $V = \bigcup_\alpha V_\alpha$ .

Formally:  $\forall x (x \neq \emptyset \implies \exists y \in x (y \cap x = \emptyset))$  — i.e.,  $y \in x$  and no  $z \in y$  also in  $x$  —  $\in$ -minimal.

Consequences:

- $x \notin x$  — else  $\{x\}$  would have no  $\in$ -minimal element — because  $x \cap \{x\} = \{x\} \neq \emptyset$ .
- No cycles  $x \in y \in x$ .
- No infinite descending  $\dots x_2 \in x_1 \in x_0$  — else  $\{x_n : n \in \omega\}$  would have no minimal.
- Every set belongs to some  $V_\alpha$  — universe well-founded — induction and recursion on  $\in$  valid — enables  $\in$ -induction: if  $\forall y \in x \varphi(y) \implies \varphi(x)$  then  $\forall x \varphi(x)$ .

Without Foundation, we could have Quine atoms  $x = \{x\}$  — consistent with other axioms — Aczel's anti-foundation studies them. Foundation rules them out — makes  $\in$  well-founded — simplifies picture — every set built from  $\emptyset$  upward.

### Choice — AC — Non-Constructive Selector

Choice function exists for any family of non-empty sets — given family  $\{X_i\}_{i \in I}$  with  $X_i \neq \emptyset$ , exists function  $f$  with  $f(i) \in X_i$  — equivalently, product  $\prod_i X_i \neq \emptyset$ , every set can be well-ordered, every vector space has basis, etc. Independent of ZF — Gödel and Cohen showed. Non-constructive, implies Banach-Tarski paradox — ball can be partitioned into finitely many pieces reassembled into two balls same size using AC.

Formally:  $\forall F [(\emptyset \notin F) \implies \exists c (c \text{ function} \wedge \text{dom}(c) = F \wedge \forall x \in F c(x) \in x)]$ .

Equivalent forms — in ZF:

- Well-Ordering Theorem — every set can be well-ordered — i.e., has order type some ordinal — implies comparability of cardinals: for any  $A, B$ ,  $|A| \leq |B|$  or  $|B| \leq |A|$ .
- Zorn's Lemma — every chain in poset has upper bound  $\rightarrow$  maximal element — used everywhere in algebra.

- Every vector space has basis — uses Zorn.
- Tychonoff — product of compact spaces compact.
- Cardinal arithmetic:  $\kappa \otimes \kappa = \kappa$  for infinite  $\kappa$ .

Why controversial historically: asserts existence of choice function without giving rule — non-constructive — for infinite family of arbitrary sets, no algorithm to pick. For finite families, provable without AC — induction. For countable families of non-empty subsets of  $\mathbb{R}$ , needs  $AC_\omega$  — weaker.

Independence: Gödel 1938 built inner model  $L$  — constructible universe — where AC holds —  $\text{Con}(\text{ZF}) \rightarrow \text{Con}(\text{ZFC})$ . Cohen 1963 forced model where AC fails — e.g.,  $\mathbb{R}$  not well-orderable —  $\text{Con}(\text{ZF}) \rightarrow \text{Con}(\text{ZF} + \neg \text{AC})$ . So AC independent.

Banach-Tarski 1924: using AC, unit ball in  $\mathbb{R}^3$  can be partitioned into 5 pieces — non-measurable — rearranged by rigid motions into two balls same size as original — paradoxical decomposition of free group  $F_2$ . Does not contradict measure — pieces non-measurable — no volume — requires AC to select representatives of orbits. Shows AC has counterintuitive geometric consequences, but no contradiction.

Together, 9 + Choice = ZFC — foundation where almost all modern mathematics lives — from  $0 = \emptyset$  to  $\mathfrak{c} = 2^{\aleph_0}$  to  $V_{\omega_1}$  — while being incomplete — CH undecidable — exactly as earlier sections on continuum.

## Key Aspects of ZFC — What It Does, Why It Matters, Where It Ends

### *Paradox Prevention — From Naive Comprehension to Size Limitation*

**Replaces unrestricted comprehension — Frege's naive  $\{x : P(x)\}$  for any  $P$  — with specific axioms like Separation and Replacement — only build subsets of already existing sets — size limitation.**

Naive set theory — Frege, Cantor 1890s — allowed: for any property  $P$ ,  $\{x : P(x)\}$  is a set. This is what Frege wrote in *Grundgesetze* — "extension of concept". Russell 1901 destroyed it with  $R = \{x : x \notin x\}$  — then  $R \in R \iff R \notin R$  — contradiction — Frege's system inconsistent.

Zermelo 1908 diagnosis: problem is *too big* comprehension — you cannot form set of all  $x$  satisfying  $P$  out of thin air. Must restrict to  $P$  inside already built set  $A$ :  $\{x \in A : P(x)\}$  — Separation.

This is **limitation of size** doctrine: a collection is a set only if not too big relative to existing universe. Russell's  $R$  is proper class — too large — not in  $V$ . Similarly class of all ordinals  $Ord$ , class of all sets  $V$  — proper classes — not sets.

How axioms enforce:

- Separation: you can only separate *within*  $A$  — if  $A$  is set,  $\{x \in A : x \notin x\}$  is just  $A$  itself minus self-containing elements — but by Foundation, no set contains itself, so it's  $A$  — no paradox.
- Replacement: image of set under function is set — if domain is set, range not too big — prevents mapping  $\omega$  onto proper class.
- Power Set + Union + Pairing: build upward gradually, not in one giant leap — each operation stays inside  $V_{\alpha+1}$  if input in  $V_\alpha$ .

Thus ZFC replaces one inconsistent axiom — unrestricted comprehension — with eight safe construction steps — Empty, Pairing, Union, Power Set, Infinity, Separation schema, Replacement schema, Foundation — each says: from sets you already have, you may form new set in controlled way. Universe  $V$  grows layer by layer  $V_0 = \emptyset$ ,  $V_{\alpha+1} = \mathcal{P}(V_\alpha)$  — cumulative hierarchy — no set contains all  $V_\alpha$  — that would be  $V$  itself — proper class — paradox avoided because  $V$  never completed as set.

This is same idea as type theory — Russell's theory of types — but implemented as size restriction, not syntactic levels.

### ***Foundation of Mathematics — Everything Is Set***

**Almost all mathematical objects — numbers, functions, topological spaces — encoded as sets within ZFC — e.g., ordered pair  $(a, b) = \{\{a\}, \{a, b\}\}$ , function as set of ordered pairs,  $\mathbb{R}$  as Dedekind cuts or Cauchy sequences —  $|\mathbb{R}|$  then defined.**

ZFC claims: mathematics = study of sets — all objects can be coded as sets — no need for urelements — primitive numbers.

Encoding chain — von Neumann construction:

- $0 := \emptyset, 1 := \{0\} = \{\emptyset\}, 2 := \{0, 1\}, n + 1 := n \cup \{n\}$  — each  $n = \{0, \dots, n - 1\}$  — ordinal  $n$  — Infinity gives  $\omega = \{0, 1, 2, \dots\} = \mathbb{N}$ .
- $\mathbb{Z}$ : equivalence classes of pairs  $(a, b) \in \mathbb{N}^2$  with  $(a, b) \sim (c, d) \iff a + d = c + b$  — intention  $a - b$  — e.g.,  $-1 = [(0, 1)]$ .
- $\mathbb{Q}$ : pairs  $(p, q) \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$  mod  $(p, q) \sim (r, s) \iff ps = rq$ .
- Ordered pair — Kuratowski 1921:  $(a, b) := \{\{a\}, \{a, b\}\}$  — property  $(a, b) = (c, d) \iff a = c \wedge b = d$  provable from Extensionality + Pairing — no extra primitive.
- Relation: subset of  $A \times B$  where  $A \times B = \{(a, b) : a \in A, b \in B\}$  — exists by Power Set + Separation —  $A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$ .
- Function: relation  $f \subseteq A \times B$  functional:  $\forall a \in A \exists! b (a, b) \in f$  — as set of ordered pairs — so function is set.
- $\mathbb{R}$  — two equivalent codings:
  - Dedekind cuts:  $r \subseteq \mathbb{Q}$  non-empty, not all  $\mathbb{Q}$ , downward closed, no maximum —  $r = \{q \in \mathbb{Q} : q < r\}$  — each real = set of rationals less than it — requires  $\mathcal{P}(\mathbb{Q})$  — Power Set.
  - Cauchy sequences: equivalence class of sequences  $(q_n)$  Cauchy — sequence is function  $\mathbb{N} \rightarrow \mathbb{Q}$  — set of pairs — exists via Power Set.
- Then  $|\mathbb{R}|$  defined: cardinality  $|X|$  = least ordinal  $\alpha$  bijective with  $X$  — exists using Choice — Well-Ordering gives alephs. So  $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0} = \mathfrak{c}$ .
- Topological space: pair  $(X, \tau)$  where  $\tau \subseteq \mathcal{P}(X)$  closed under unions and finite intersections —  $\tau$  exists as subset of  $\mathcal{P}(X)$  — Power Set twice.
- Group: pair  $(G, \cdot)$  where  $\cdot : G \times G \rightarrow G$  function — etc.

Thus: number theory, analysis, algebra, topology all become theorems about  $V$  — metamathematical reduction — Hilbert's program — shows consistency of mathematics reduces to consistency of ZFC — though Gödel's second incompleteness says ZFC cannot prove its own consistency.

Practical upshot: when mathematician says "let  $f : \mathbb{R} \rightarrow \mathbb{R}$  continuous", ZFC guarantees  $f$  exists as set in  $V_{\omega+2}$  — Power Set twice over  $\omega$ .

### ***Independence — Where ZFC Ends — Continuum Hypothesis as Test Case***

**Foundational results, such as independence of continuum hypothesis, studied within this framework — CH is statement about cardinal arithmetic  $2^{\aleph_0}$  — cannot be proved or refuted in ZFC — shows ZFC incomplete — Gödel incompleteness — need new axioms — large cardinals, forcing axioms like Martin's Maximum — which decide CH**



**one way or other — e.g.,  $PFA \implies \mathfrak{c} = \aleph_2$  — active research — Woodin's Ultimate-L program aims to find canonical universe where CH false.**

*What is CH?* Cantor:  $|\mathbb{N}| = \aleph_0 < |\mathcal{P}(\mathbb{N})| = \mathfrak{c} = 2^{\aleph_0}$ . CH:  $\mathfrak{c} = \aleph_1$  — smallest uncountable cardinal — no cardinal strictly between  $\aleph_0$  and  $\mathfrak{c}$ . Generalized CH:  $2^{\aleph_\alpha} = \aleph_{\alpha+1}$ .

ZFC proves  $2^{\aleph_0} \geq \aleph_1$ , but cannot prove equality or inequality — independent.

*Gödel 1938 — CH consistent:* Construct inner model  $L$  — constructible universe —  $L_0 = \emptyset$ ,  $L_{\alpha+1} = \text{Def}(L_\alpha)$  — definable subsets only, not all subsets —  $L = \bigcup L_\alpha$ .  $L \models ZFC + CH$  — indeed  $L \models V = L$  and  $GCH$  — and  $L \subseteq V$  with same ordinals — so if ZF consistent, ZFC+CH consistent — CH cannot be refuted — no proof of  $\neg CH$  in ZFC.

*Cohen 1963 —  $\neg CH$  consistent:* Forcing — start with countable transitive model  $M \models ZFC$ , add new reals — generic filter  $G$  over poset  $\text{Add}(\omega, \aleph_2)$  — adds  $\aleph_2$  many Cohen reals —  $M[G] \models ZFC + 2^{\aleph_0} \geq \aleph_2$  — so  $\neg CH$  holds — and  $M$  has same ordinals as  $M$  — so if ZF consistent, ZFC+ $\neg CH$  consistent — CH cannot be proved.[G]

Together: CH independent — first major independence after Gödel.

*Philosophical consequence — incompleteness:* Gödel's first incompleteness: any recursive consistent extension of ZF is incomplete — there are statements undecidable. CH is natural example — not artificial self-reference, but central cardinal arithmetic.

Thus ZFC does not decide size of continuum —  $\mathfrak{c} = 2^{\aleph_0}$  is definite size — bigger than countable — proved  $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$  — but its exact place in aleph hierarchy  $\aleph_1, \aleph_2, \dots$  undetermined.

Modern program to decide:

- Large cardinals: axioms asserting existence of huge cardinals — inaccessible, measurable, supercompact — with elementary embeddings  $j : V \rightarrow M$  — imply more  $V$  tall — decide many low-level statements — e.g., measurable implies  $V \neq L$  — but do not decide CH — Levy-Solovay theorem.
- Forcing axioms — Martin's Maximum MM, Proper Forcing Axiom PFA: assert that any forcing notion with certain preservation property that could make a statement true already has made it true — maximize universe width — imply  $2^{\aleph_0} = \aleph_2$  — so  $PFA \implies \mathfrak{c} = \aleph_2$  — CH false — and  $\aleph_2$  has rich combinatorics — e.g., all Aronszajn trees special. So CH false under forcing axioms — favoured by many set theorists — e.g., Foreman-Magidor-Shelah.
- Ultimate-L — Woodin: try to build canonical inner model that can accommodate supercompact cardinal — like  $L$  but bigger — *Ultimate* —  $L \models CH$  — predicts CH true —  $V = \text{Ultimate} - L$  would imply  $GCH$  — opposite direction.
- Inner model program vs forcing axioms = current schism — whether to maximize or canonicalize  $V$ .

**Philosophical upshot:  $\mathfrak{c} = 2^{\aleph_0}$  is definite size — bigger than countable — but its exact place in aleph hierarchy undetermined by current axioms — continuum's "true" cardinality is like parallel postulate — may depend on extra axioms we choose — Platonists believe one true answer, formalists accept many models.**

Analogy exact: Euclid's parallel postulate independent of other axioms — leads to Euclidean vs hyperbolic geometry — many models. Similarly CH independent of ZFC — leads to  $L$  where CH true, forcing extensions where CH false — many models  $V$  — multiverse — Hamkins — vs universe view — Woodin — one true  $V$  we have not axiomatized yet.

Platonist:  $V$  real —  $2^{\aleph_0}$  has true value — either  $\aleph_1$  or  $\aleph_2$  or larger — ZFC just too weak to see it — need new true axioms — e.g., large cardinals.

Formalists / multiversist: independence means CH has no absolute truth value — like "is  $\theta$  parallel?" without specifying geometry — there are many set-theoretic universes, some CH, some  $\neg$ CH, equally legitimate — we study all.

For working mathematician: ZFC suffices — almost all theorems of analysis, algebra, number theory provable in ZFC — CH irrelevant to most — but for set theory, topology, infinite combinatorics — e.g., existence of Suslin tree, Whitehead problem — CH matters — independence shows limit of current foundation — active research on new axioms deciding continuum.

## Iwasawa Theory — Arithmetic Over Infinite Towers

Initiated by Kenkichi Iwasawa in late 1950s — papers 1959, 1969 — Iwasawa theory is program that studies growth of arithmetic objects — class groups, Selmer groups — in infinite  $p$ -adic towers of number fields, by packaging them into modules over  $p$ -adic power series ring, and connects these algebraic objects to  $p$ -adic  $L$ -functions via "main conjecture," proved by Mazur & Wiles 1984 — one of deepest links between algebra and analysis in number theory —  $p$ -adic analogue of analytic class number formula, precursor to modern  $p$ -adic Langlands.

Classical number theory studies single field  $F$  — e.g.,  $\mathbb{Q}(\sqrt{-5})$  class number 2. Iwasawa's insight: instead of single field, study whole tower  $F_\infty/F$  with Galois group  $\mathbb{Z}_p$  —  $p$ -adic direction — and asymptotic behavior often simpler, controlled by  $p$ -adic analytic function.

### Key Aspects of Iwasawa Theory

- **Infinite Towers —  $\mathbb{Z}_p$ -extensions:** Theory looks at tower of number fields  $F = F_0 \subset F_1 \subset F_2 \subset \dots \subset F_\infty = \bigcup F_n$ , where Galois group  $\text{Gal}(F_n/F)$  is cyclic of order  $p^n$ , and total Galois group  $\Gamma = \text{Gal}(F_\infty/F) \cong \mathbb{Z}_p$  — additive group of  $p$ -adic integers — topological generator  $\gamma$ . So  $\Gamma \cong \varprojlim \mathbb{Z}/p^n\mathbb{Z}$  — profinite.

Archetypal example: cyclotomic  $\mathbb{Z}_p$ -extension of  $\mathbb{Q}$ :  $F_n = \mathbb{Q}(\mu_{p^{n+1}})^+$ ? Actually cyclotomic:  $\mathbb{Q}_\infty = \bigcup \mathbb{Q}(\mu_{p^n})$  —  $F_n = \mathbb{Q}(\mu_{p^n})$ , Galois  $(\mathbb{Z}/p^n)^\times \cong (\mathbb{Z}/p)^\times \times \mathbb{Z}/p^{n-1}$  —  $\mathbb{Z}_p$ -part. For  $F = \mathbb{Q}$ ,  $F_n = \mathbb{Q}(\zeta_{p^n})$ ,  $\text{Gal}(F_\infty/F) \cong \mathbb{Z}_p^\times \cong \Delta \times \Gamma$ ,  $\Delta = (\mathbb{Z}/p)^\times$  finite,  $\Gamma \cong \mathbb{Z}_p$ . So tower ramified only at  $p$  — totally ramified. Also anti-cyclotomic extensions for imaginary quadratic fields.

Why  $\mathbb{Z}_p$ ? Because  $p$ -adic Lie group of dimension 1 — Iwasawa algebra manageable — power series.

- **Growth of Class Groups — Iwasawa's Formula:** Iwasawa 1959 showed  $p$ -part of class number  $h_{F_n}$  — size of ideal class group  $\text{Cl}_{F_n}$  — follows strict formula for large  $n$ :

Write  $|\text{Cl}_{F_n}[p^\infty]| = p^{e_n}$  —  $p$ -primary part. Then exists  $n_0$  such that for  $n \geq n_0$ :

$$e_n = \mu p^n + \lambda n + \nu$$

with  $\mu, \lambda, \nu$  constants — Iwasawa invariants —  $\mu, \lambda \geq 0$  integers,  $\nu$  integer — independent of  $n$ .

Proof idea:  $X_\infty = \varprojlim \text{Cl}_{F_n}[p^\infty]$  — inverse limit via norms — is finitely generated torsion module over Iwasawa algebra  $\Lambda$ . Structure theorem for  $\Lambda$ -modules gives invariants  $\mu, \lambda$ . Growth formula follows from  $\Lambda$ -module size calculation —  $e_n = \mu p^n + \lambda n + \nu$  essentially  $\Lambda/(p^\mu, f(T))$  growth.

Interpretation:  $\mu$  measures exponential growth —  $p$ -torsion wild —  $\lambda$  measures linear growth —  $\mathbb{Z}_p$ -rank. Iwasawa conjectured  $\mu = 0$  for cyclotomic  $\mathbb{Z}_p$ -extension of any number field — proved by Ferrero-Washington 1979 for  $F/\mathbb{Q}$  abelian —  $F$  abelian over  $\mathbb{Q}$  then  $\mu = 0$ . For general  $F$ , still open.  $\lambda$  can be large.

Example: For  $F = \mathbb{Q}$ , cyclotomic tower, class groups  $p$ -parts often stable —  $\mu = \lambda = 0$  for regular primes  $p$  — Kummer. For irregular primes,  $\lambda > 0$  relates to  $p$  dividing Bernoulli numbers.

- **Iwasawa Algebra:** Study involves Iwasawa algebra  $\Lambda = \mathbb{Z}_p[[\Gamma]]$  — completed group ring — inverse limit  $\varprojlim \mathbb{Z}_p[\Gamma/\Gamma^{p^n}]$  — isomorphic to power series ring  $\mathbb{Z}_p[[T]]$  via  $\gamma \mapsto 1 + T$  — isomorphism  $\Lambda \cong \mathbb{Z}_p[[T]]$  — choose topological generator  $\gamma$  of  $\Gamma$ , send  $\gamma - 1 \mapsto T$ . So  $\Lambda$  is 2-dimensional regular local ring, UFD, complete.

Arithmetic objects like inverse limit of class groups  $X_\infty = \varprojlim \text{Cl}_{F_n}[p^\infty]$  are modules over  $\Lambda$  — action of  $\Gamma$  via Galois — finitely generated torsion. Structure theorem: pseudo-isomorphic — map with finite kernel and cokernel — to

$$X_\infty \sim \bigoplus_i \Lambda/p^{\mu_i} \oplus \bigoplus_j \Lambda/f_j(T)^{n_j}$$

where  $f_j$  distinguished polynomials — monic, all non-leading coefficients divisible by  $p$ . Then  $\mu = \sum \mu_i$ ,  $\lambda = \sum n_j \deg f_j$ . Characteristic ideal  $\text{char}_\Lambda(X_\infty) = (p^\mu \prod f_j^{n_j})$  — principal — encodes invariants.

- **Main Conjecture — Algebraic = Analytic:** Proved by Barry Mazur and Andrew Wiles 1984 for  $F = \mathbb{Q}$ , and Wiles 1990 for totally real fields — this conjecture bridges algebraic objects and analytic functions, stating that characteristic ideal of Iwasawa module is generated by  $p$ -adic  $L$ -function.

Classical side: Kubota-Leopoldt  $p$ -adic  $L$ -function  $L_p(s, \chi)$  —  $p$ -adic analytic function interpolating values  $L(1 - n, \chi) = -B_{n, \chi}/n$  for  $n \geq 1$  —  $B_{n, \chi}$  generalized Bernoulli numbers. For even Dirichlet character  $\chi$ , there exists  $p$ -adic  $L$ -function  $L_p(s, \chi)$  with  $L_p(1 - n, \chi) = (1 - \chi\omega^{-n}(p)p^{n-1})L(1 - n, \chi\omega^{-n})$ .

Algebraic side:  $X_\infty^\chi$  —  $\chi$ -isotypic component of  $X_\infty$  — inverse limit of  $p$ -class groups in cyclotomic tower.

Main Conjecture:

$$\text{char}_\Lambda(X_\infty^\chi) = (L_p(s, \chi))$$

as ideals in  $\Lambda$  — after identifying  $T = \gamma^s - 1$ ? More precisely, for each even  $\chi$ , there is power series  $G_\chi(T) \in \Lambda$  such that  $L_p(s, \chi) = G_\chi(u^s - 1)/H_\chi$  and  $\text{char} = (G_\chi)$ . So zeros of  $p$ -adic  $L$ -function control sizes of class groups —  $p$ -adic class number formula in family.

Wiles proof used Eisenstein ideal on Hilbert modular varieties — constructed unramified extension via Galois representations attached to Eisenstein series — proved divisibility both ways — analytic | algebraic via congruences, algebraic | analytic via class field theory + Ribet's converse to Herbrand.

Corollary: implies Kummer-Vandiver, Iwasawa's  $\mu = 0$  consequences, and main ingredient for proof of Bloch-Kato, BSD, and later for Wiles' proof of Fermat via modularity? Not directly, but techniques overlapping — Galois representations.

- **Iwasawa Main Conjectures — Elliptic Curves and Higher Dimensions:** Generalizations of main conjecture exist for elliptic curves and higher-dimensional varieties, linking Selmer groups to  $p$ -adic  $L$ -functions, as highlighted in studies on link to Springer Book on Iwasawa Theory 2012 and research on Elliptic Curves and Iwasawa's  $\mu = 0$  Conjecture.

For elliptic curve  $E/\mathbb{Q}$  and  $p$  prime, consider cyclotomic tower  $\mathbb{Q}_\infty$ . Define Selmer group  $\text{Sel}_{p^\infty}(E/F_n)$  — measures  $p$ -adic Tate-Shafarevich and rational points — inverse limit  $X_E = \varprojlim \text{Sel}^\vee$  dual —  $\Lambda$ -module. Mazur's Control Theorem shows it is  $\Lambda$ -torsion if  $E$  has good ordinary reduction at  $p$ .

Conjecture — Mazur, Greenberg 1970s:  $\text{char}_\Lambda(X_E) = (L_p(E, T))$  where  $L_p(E, T)$  is Mazur-Swinnerton-Dyer  $p$ -adic  $L$ -function interpolating  $L(E, \chi, 1)/\Omega_E$ . Proved for many cases: Kato 2004 proves divisibility, Skinner-Urban 2014 proves equality for many ordinary  $E$  when  $p$  split multiplicative? — big result.

For supersingular  $p - a_p = 0$  —  $X_E$  not torsion — need signed Selmer groups — Kobayashi, Pollack — plus/ minus  $p$ -adic  $L$ -functions — Sprung.

Also anti-cyclotomic theory — Heegner points — Bertolini-Darmon.

For modular forms, Galois representations — main conjectures formulated by Greenberg, Perrin-Riou, Kato — linking motivic Selmer groups to  $p$ -adic  $L$ -functions — now in vast framework of Euler systems — Kato's Euler system, Kolyvagin's.

Iwasawa's  $\mu = 0$  Conjecture for elliptic curves:  $X_E$  has  $\mu = 0$  — conjectured by Coates-Sujatha, Greenberg — still open generally.

In summary: Iwasawa theory transforms intractable growth problem — how class numbers vary in tower — into structure theory of modules over  $\mathbb{Z}_p[[T]]$  — then into  $p$ -adic analysis via main conjecture \$algebraic characteristic ideal = analytic  $p$ -adic  $L$ -function\$ — algebra = analysis —  $p$ -adic analogue of analytic class number formula  $\zeta_F^* \sim hR$ . Modern number theory: one of main tools to approach BSD — Birch Swinnerton-Dyer — via Iwasawa Main Conjecture for elliptic curves.